

Disorder-Induced Phase Transitions in Altermagnetic Josephson Junctions

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Altermagnetic Josephson junctions (AMJJs) can host unconventional π phase and φ phase despite vanishing net magnetizations. Whether these phases are stable against disorder existing in real materials remains an open question. Here, we investigate impact of disorder on exotic phases in two-dimensional d -wave AMJJs. We show that disorder is able to induce phase transitions between the exotic π and conventional 0 phases, accompanied by a strong suppression of critical current. This behavior is attributed to modifications of the tunneling Cooper-pair phase shift and superconducting decoherence. Remarkably, the anomalous φ phase is highly fragile in presence of disorder and can be driven to either a π phase or 0 phase in a nonreciprocal manner. Across such transitions, the first harmonic of current-phase relation changes sign, while the higher-order harmonics are rapidly suppressed. Our findings reveal the crucial role of disorder in tailoring distinct phases of AMJJs and shed new light on their potential functionalities.

Introduction.—Altermagnets (AMs) have emerged as a new class of magnetic phase characterized by momentum-dependent spin splitting but with vanishing net magnetization [1–10]. The AMs combine key features of conventional ferromagnets with spin-split band structures and antiferromagnets with net zero magnetization, thereby providing a fertile platform for novel physics and potential applications [11–36]. In particular, the vanishing net magnetization eliminates the detrimental stray fields when proximitized to superconductors (SCs) [37–39]. Henceforth, the hybrid structure of AM-SC provides promising functionalities [40–44]. A prominent example is the altermagnetic Josephson junction (AMJJ) [45–54]. With d -wave AMs, the AMJJ demonstrates tunable 0- π phase oscillations as a function of the junction length [45–48], analogous to those previously studied in ferromagnetic or antiferromagnetic Josephson junctions with finite magnetizations [55–58]. Such controllable π -phase Josephson junctions are of particular interest for superconducting qubits and digital circuits [59–61]. In addition, AMJJs can host anomalous φ phase with supercurrent nodal points residing at general phase differences $\phi \neq 0, \pi$ [62]. However, to construct AMJJs with the confirmed AM materials, such as $\text{KV}_2\text{Se}_2\text{O}$ [63–65] and CrSb [66–69], the presence of disorder will significantly modify their quantum transport properties [36]. Henceforth, a crucial open question is how will these exotic phases of AMJJs be influenced by disorder? Addressing this problem will not only reveal the unique interplay between AMs, disorder, and superconductivity, but also provide useful guidance for practical implementations of AM-based Josephson devices.

In this work, we systematically investigate the impact of disorder on exotic phases of d -wave AMJJs in two dimensions (2D). The AMJJ is consisted of an SC-AM hybrid structure as shown in Fig. 1(a), where disorder is

introduced in the altermagnetic junction region. We find that the exotic π phase in AMJJs is sensitive to disorder and can be driven to the conventional 0 phase, even at a relatively weak disorder strength. The corresponding critical supercurrent will be strongly suppressed. Disorder can also induce a transition from the conventional 0 phase to the exotic π phase in the reverse direction, making 0- π phase transition reciprocally possible [see Fig. 1(b)]. Remarkably, the anomalous φ phase turns out to be highly fragile against disorder. It is eventually driven to either a π phase or 0 phase in a non-reciprocal manner. These results hold generally for different altermagnetic orientations and junction details. We demonstrate strong numerical evidence for these phenomena and provide direct interpretations based on disorder-induced phase shift of Cooper pairs tunneling across the junction.

Model and symmetry analysis.—We consider a disordered AMJJ schematically illustrated in Fig. 1(a). The system consists of two superconductor leads ($j < 0$ and $j > L_x + 1$) separated by a disordered d -wave AMs (i.e., $1 \leq j \leq L_x$). At the interfaces, thin normal-metal barrier layers are introduced with a control potential V_b . The lattice sites can be labeled by the basis vector $\mathbf{r} = j\mathbf{e}_x + k\mathbf{e}_y$, where $\mathbf{e}_x$ and $\mathbf{e}_y$ are unit vectors in x and y directions, respectively [70]. Periodic boundary conditions are imposed along y direction. The mean-field tight-binding Hamiltonian reads

$$\mathcal{H}(\phi) = \sum_{\mathbf{r}, \mathbf{r}'} [\Psi_{\mathbf{r}}^\dagger h_{\mathbf{r}, \mathbf{r}'} \Psi_{\mathbf{r}'} - \bar{\Psi}_{\mathbf{r}} h_{\mathbf{r}, \mathbf{r}'}^* \bar{\Psi}_{\mathbf{r}'}^\dagger] + \sum_{\mathbf{r} \in \text{SCs}} [\Psi_{\mathbf{r}}^\dagger \Delta_{\mathbf{r}} \bar{\Psi}_{\mathbf{r}}^\dagger - \bar{\Psi}_{\mathbf{r}} \Delta_{\mathbf{r}}^* \Psi_{\mathbf{r}}], \quad (1)$$

where $\Psi_{\mathbf{r}} = (C_{\mathbf{r}\uparrow}, C_{\mathbf{r}\downarrow})^T$ with $C_{\mathbf{r}\sigma}^\dagger$ ($C_{\mathbf{r}\sigma}$) the creation (annihilation) operator for an electron at site $\mathbf{r}$ of spin $\sigma = \{\uparrow, \downarrow\}$. $\bar{\Psi}_{\mathbf{r}}$ means the transpose of $\Psi_{\mathbf{r}}$. The spin-singlet pairing potential takes $\Delta_{\mathbf{r} \in L} = i\sigma_y \Delta$ and

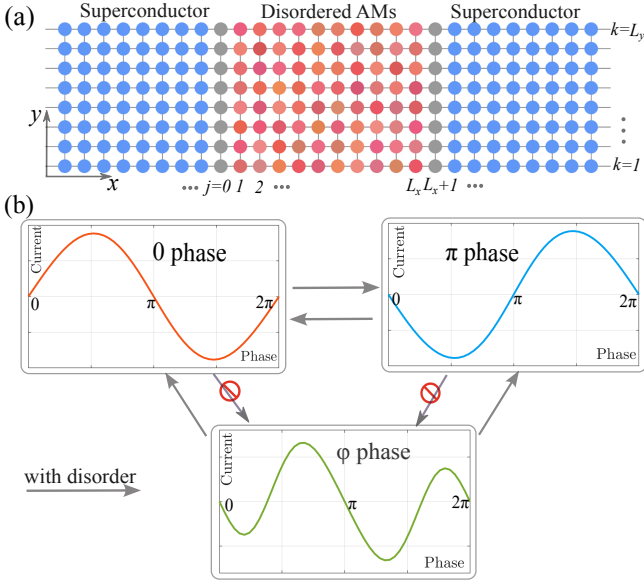


FIG. 1. (a) Schematic of the disordered AMJJ in 2D. Two s -wave superconductor leads (blue color) with a phase difference ϕ are connected by a disordered d -wave AM ($d_{x^2-y^2}$ -wave or d_{xy} -wave) with a length of L_x (red color). The color variation indicates random onsite potentials at different lattice sites. There are normal metals at two interfaces with a barrier potential of V_b (gray color). Periodic boundary conditions are applied along y direction. (b) Sketch of the disorder-induced phase transitions between different phases of the AMJJ. Note that the transitions between 0 phase (red color) and π phase (blue color) are reciprocal, whereas the transition from φ phase (green color) to either 0 phase or π phase is nonreciprocal.

$\Delta_{\mathbf{r} \in R} = i\sigma_y \Delta e^{i\phi}$ with a phase difference ϕ across the left (L) and the right (R) lead. Explicitly, the normal-state Hamiltonian $h_{\mathbf{r},\mathbf{r}'}$ containing the $d_{x^2-y^2}$ -wave AMs junction region on a square lattice is given by [6, 7]

$$h_{\mathbf{r},\mathbf{r}'} = \sum_{\sigma,\sigma'} \{-t\delta_{|\mathbf{r}-\mathbf{r}'|,1} - t_{J,\mathbf{r}}[\mathbf{e}_{\langle\mathbf{r}\mathbf{r}'\rangle} \cdot \boldsymbol{\sigma}]_{\sigma\sigma'} - \mu + V_{\mathbf{r}}\delta_{\mathbf{r},\mathbf{r}'}\}, \quad (2)$$

where t is the nearest-neighbor hopping amplitude, μ is the chemical potential, and $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ denotes the Pauli matrix vector for spin degrees of freedom. The spin-dependent altermagnetic hopping amplitude takes $t_{J,\mathbf{r}} = t_J$ in the AM region and vanishes elsewhere. The unit vector $\mathbf{e}_{\langle\mathbf{r}\mathbf{r}'\rangle}$ for nearest-neighbor hopping in AMs is set as $\mathbf{e}_{\langle\mathbf{r}\mathbf{r}'\rangle} = +\mathbf{e}_z$ along x direction and $\mathbf{e}_{\langle\mathbf{r}\mathbf{r}'\rangle} = -\mathbf{e}_z$ along y direction, respectively. The clean AM is described by an effective Hamiltonian $H_0(\mathbf{k}) = -(2t \cos k_x + 2t \cos k_y + \mu) - 2t_J(\cos k_x - \cos k_y)\sigma_z$, where $\mathbf{k} = (k_x, k_y)$ is the momentum vector. Disorder is introduced in the AM region via a random on-site potential $V_{\mathbf{r}}$ drawn randomly from the interval $[-W/2, W/2]$ with W being the disorder strength. At the interface layers of normal metals, the onsite potential is taken by a constant as $V_{\mathbf{r}} = V_b$, while

it vanishes in the superconducting leads.

The supercurrent across the junction is obtained using the recursive Green's function as [71–73]

$$I(\phi) = \frac{ieT}{\hbar} \sum_{\omega_n} \text{Tr}\{\tau_z H_{j,j+1} \hat{G}_{j+1,j}(i\omega_n) - \tau_z H_{j+1,j} \hat{G}_{j,j+1}(i\omega_n)\}, \quad (3)$$

where T is the temperature, $H_{j,j+1}$ is the hopping matrix between two adjacent j - and $(j+1)$ -th lattice layers in the middle junction part, and $\hat{G}_{j,j+1}(i\omega_n)$ is the corresponding Nambu Green's function evaluated at Matsubara frequency $\omega_n = (2n+1)\pi/\beta$ with $\beta = 1/k_B T$. Here, n is an integer number, k_B is the Boltzmann constant, and τ_z the Pauli matrix in Nambu space. The temperature dependence of superconducting gap is taken as $\Delta(T) = \Delta_0 \tanh[1.74\sqrt{T_c/T} - 1]$, where $\Delta_0 = 1.76k_B T_c$ and T_c is the critical temperature.

The symmetry properties of AMJJs impose strong constraint on the current-phase relation (CPR). The AM breaks time-reversal symmetry $\mathcal{T}$ and four-fold rotation symmetry C_4 , while it respects the combined symmetry $C_4\mathcal{T}$ [74]. For AMJJs, this particular symmetry implies $[C_4\mathcal{T}]\mathcal{H}(\phi)[C_4\mathcal{T}]^{-1} = \mathcal{H}(-\phi)$ [48, 62]. Correspondingly, the free energy, i.e. $F(\phi) = -\frac{\ln Z(\phi)}{\beta}$ with $Z(\phi) = \text{Tr}[e^{-\beta\mathcal{H}(\phi)}]$, follows a relation $F(\phi) = F(-\phi)$. The supercurrent can be derived as $I(\phi) = \partial_\phi F(\phi)$, yielding a constrained CPR as $I(\phi) = -I(-\phi)$. Therefore, the allowed phases in AMJJs are just 0 phase, π phase, and φ phase. In the following, we investigate disorder-induced phase transitions among these phases.

Transitions between π phase and 0 phase.—We first examine the fate of exotic π phase in presence of disorder, as illustrated in Figs. 2(a) and 2(e). Specifically, Fig. 2(a) displays the normalized CPR $I(W, \phi)/I_c(W)$ as a function of disorder strength W . At $W = 0$, the AMJJ resides in the π phase [75]. With increasing W , the system undergoes a transition to the conventional 0 phase at a critical disorder strength $W_{c1} \approx 1.2t$. The corresponding critical current is plotted in Fig. 2(e). Across the transition, I_c changes sign, consistent with the π to 0 phase transition. In addition, its magnitude exhibits a dip near W_{c1} and is significantly suppressed as disorder strength further increases. These results indicate that the exotic π phase is not robust against disorder whereas can be driven into a conventional 0 phase even at a relatively weak disorder strength.

With further increasing W , the supercurrent is strongly suppressed [see Fig. 2(e)]. For $W_c > 2.5t$, the critical current is reduced to an order of $I_c(W)/I_c(0) \sim 10^{-3}$. In this regime, the strong fluctuation of supercurrent due to disorder renders the CPR ill-defined even after averaging over 5×10^3 configurations. We therefore exclude this strongly fluctuating regime from further

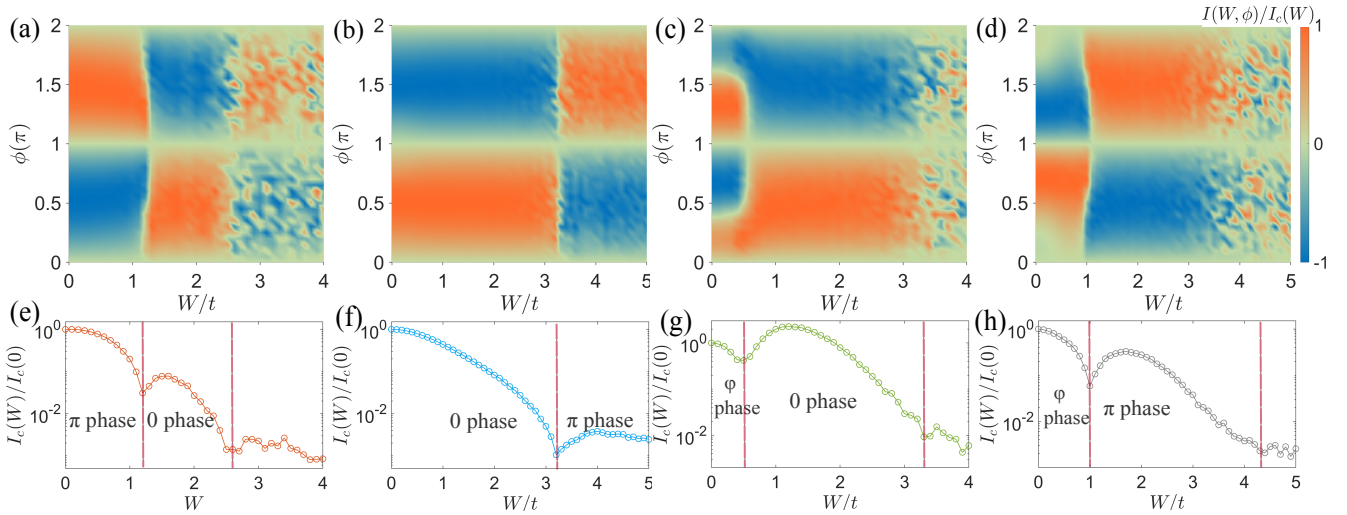


FIG. 2. (a-d) Evolution of the CPR with increasing disorder strength W . In the clean limit with $W = 0t$, the AMJJ is in π phase for (a) with $L_x = 16$, 0 phase for (b) with $L_x = 8$, φ phase in (c) with $L_x = 18$, and φ phase in (d) with $L_x = 14$, respectively. (e-h) The critical supercurrent as a function of W corresponding to panels (a-d). Other parameters are $L_y = 30$, $V_b = 2t$, $t_J = 0.3t$, $\Delta = 0.1t$, $T = 0.05T_c$, and $\mu = -2t$. We have taken 5×10^3 disorder configurations for all plots.

analysis [76] and focus on the phase transition associated with the well-defined CPRs.

Notably, disorder can also induce a transition from the conventional 0 phase to the exotic π phase in a reversal direction. This behavior is demonstrated in Figs. 2(b) and 2(f). In the clean limit, the AMJJ resides in the 0 phase. As increasing W , the critical current decays rapidly until the phase transition point at around $W_c \approx 3.2t$. Upon further increasing W , the supercurrent changes its sign, indicating a transition into the π phase. After the phase transition, the critical current is reduced to an order of $I_c(W)/I_c(0) \sim 10^{-2}$, while the CPR remains well-defined.

As we have shown, the impact of disorder in AMJJs manifests in two main aspects: (i) it drives exotic phase transitions; (ii) it suppresses the critical supercurrent. We elucidate these two points as follows. In AMJJs, the Cooper pairs tunneling through the junction will acquire a finite momentum $\delta q \propto t_J$ even there is no net magnetization [45, 47]. The supercurrent across the junction of length L_x is thus determined by an extra phase factor $\delta q L_x$ as $I \propto \cos(\delta q L_x)$ [47, 56]. Disorder modifies this behavior by effectively reducing the altermagnetic spin-split strength, $t_J \rightarrow t'_J$, due to the smearing of the anisotropic spin-dependent band structure [36, 77]. This thus leads to a renormalized phase shift to $\delta q' L$. Therefore, the critical current can flip its sign with changing the phase shift, enabling the transition between the 0 and π phases. Note that the scattering from disorder is detrimental to superconducting coherence [71]. The strong disorder destroys the coherence of tunneling Cooper pairs such that the critical current is substantially suppressed, giving rise to the strongly fluctuating regime.

Fragile φ phase.—AMJJs can host the anomalous φ phase, which arises when higher-order harmonics $I_n \sin(n\phi)$ (with $n \geq 2$) dominates the CPR [45, 62]. In this case, the supercurrent can take zero values at $\phi \neq 0, \pi$. The φ phase, however, turns out to be highly fragile under disorder. We illustrate this result in Figs. 2(c-d) and 2(g-h). The system is initially in a φ phase in the clean limit. Upon introducing disorders, even at weak strength, it is driven into either the 0 phase [Fig. 2(c)] or the π phase [Fig. 2(d)], depending on initial conditions. The critical current drops at the critical points, signaling a phase transition. Surprisingly, the critical current can even be slightly enhanced after the phase transition [Fig. 3(g)]. The typical CPRs before and after phase transitions are presented in Figs. 3(a) and 3(b), respectively. Clearly, the induced 0 phase and π phase after phase transitions exhibit smooth and sinusoidal CPRs.

The appearance of φ phase is dominated by higher-order harmonics in the CPRs [62]. Thus we explain the fragility of the φ phase in terms of suppression of higher-order harmonics by disorder. To this end, we analyze the harmonic content via Fourier decomposition of the CPR as

$$I(W, \phi) = \sum_{n=1}^{\infty} I_{nh}(W) \sin(n\phi), \quad (4)$$

where $I_{nh}(W)$ is the amplitude of n -th order harmonic. We perform fast Fourier transformation of CPRs corresponding to transitions of φ phase in Figs. 2(c) and 2(d). We illustrate this point in Fig. 3(c) as an example. The three lowest orders with $n \leq 3$ are plotted as a function of disorder strength W . In the clean limit, the second harmonic I_{2h} is comparable to first harmonic I_{1h} , while

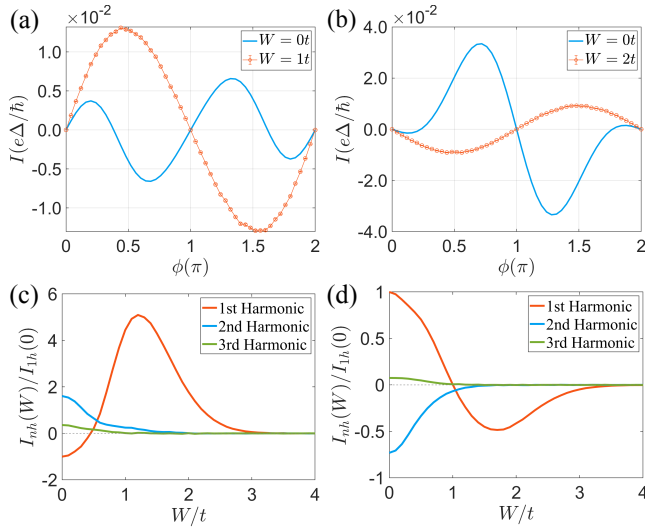


FIG. 3. (a-b) CPRs of the AMJJ at different disorder strength W . Panel (a) demonstrates φ phase before the transition and 0 phase after the transition. The AM junction length is of $L_x = 18$. Panel (b) demonstrates φ phase before transition and π phase after transition. The AM junction length is of $L_x = 14$. (c-d) Corresponding harmonic components in CPRs as a function of W , corresponding to Fig. 2(c) and 2(d), respectively. Other parameters are the same as in Fig. 2.

higher-order contributions are negligible. As increasing the disorder strength W , I_{2h} and I_{3h} drop to zero in a monotonic way. In contrast, the first harmonic I_{1h} exhibits a non-monotonic behavior: its magnitude decays to zero first, then changes sign, and subsequently increases to a maximum before being totally suppressed by strong disorder. As a result, the φ phase collapses once the higher-order harmonics are fully suppressed, leaving a CPR dominated by the first harmonic. This behavior can also be understood from the free energy point of view. Note that the CPR can be derived from $I(\phi) = \partial_\phi F(\phi)$. The φ phase corresponds to local extrema of free energy function $F(\phi)$ at $\phi \neq 0, \pi$ [62]. These local extrema are not protected by any symmetries, unlike the global ones at $\phi = 0/\pi$. Random potentials smear out these features, stabilizing only the global extrema at $\phi = 0/\pi$, thereby suppressing higher-order harmonics and destroying the φ phase. Consequently, disorder cannot induce a transition from the π phase and 0 phase back to the φ phase in a reversal direction, consistent with Fig. 2. Note that the sign of the first harmonic determines whether the φ phase is driven to a 0 phase or a π phase.

Dependence on junction width and temperature.—To demonstrate the generality of disorder-induced phase transitions in AMJJs, we examine the dependence on junction width and temperature. We illustrate this point by taking the transitions between 0 phase and π phase as an example, as shown in Fig. 4. Figures 4(a) and 4(b) display the critical current $I_c(W)/I_c(0)$ as a function of

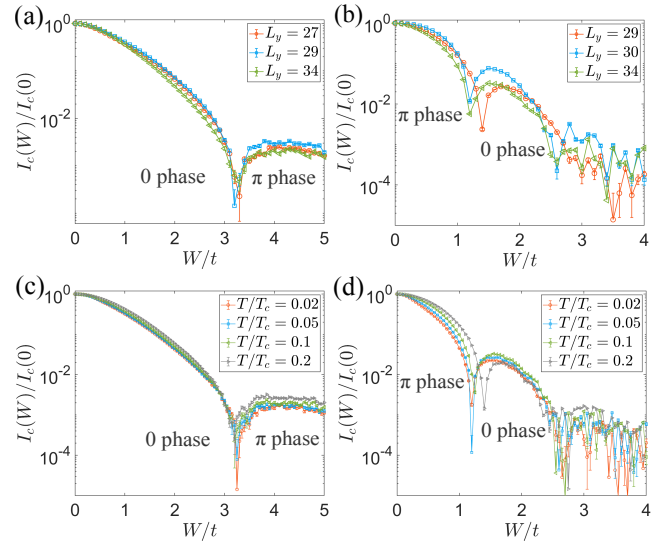


FIG. 4. (a,b) The width dependence of critical supercurrent as a function of disorder strength W . The temperature is fixed at $T = 0.05T_c$ and the junction length is $L_x = 8$. (c,d) The temperature dependence of critical supercurrent as a function of disorder strength W . The width of the junction is fixed at $L_y = 34$ and the junction length is $L_x = 16$. We have taken 1×10^4 disorder configurations for all plots. Other parameters are $V_b = 2t$, $t_J = 0.3t$, $\Delta = 0.1t$, and $\mu = -2t$.

W for different junction width L_y . The transitions between the 0 phase and π phase show robustness against small changes of the width L_y , whereas the critical disorder exhibits slight shift. The temperature dependence is presented in Figs. 4(c) and 4(d). Note that we follow the same form of $\Delta(T) = \Delta_0 \tanh[1.74\sqrt{T_c/T} - 1]$ as before. With increasing W , the critical current decays rapidly prior to the phase transition. This decay is more pronounced at lower temperatures, indicating the stronger influence of disorder effect on the superconducting transport. The phase transition persists at different temperatures, while the corresponding critical disorder strength is slightly modified. At strong disorder regime in Figs. 4(b) and 4(d), the critical current demonstrates strong fluctuations corresponding Fig. 2(a).

The d_{xy} -wave AMs case.—The anisotropy of AM orders has been found to affect the Josephson effect significantly [45–48]. Therefore, we consider disorder-induced phase transitions in d_{xy} -wave AMJJs, as a comparison to the $d_{x^2-y^2}$ -wave case. In the clean limit, the lattice model can be obtained by discretizing the effective Hamiltonian [48] $H_1(\mathbf{k}) = -(2t \cos k_x + 2t \cos k_y + \mu)\sigma_0 + 2t_J \sin k_x \sin k_y \sigma_z$ [78]. The random on-site potential V_r is added in the altermagnetic junction as described above. Figure 5 presents the main results for disorder-induced phase transitions in d_{xy} -wave AMJJs. As shown in Figs. 5(a) and 5(c), an initial π phase is first driven to a 0 phase at a critical disorder strength $W_c \approx 1.3t$. Interestingly, upon further increasing W , the system reenters

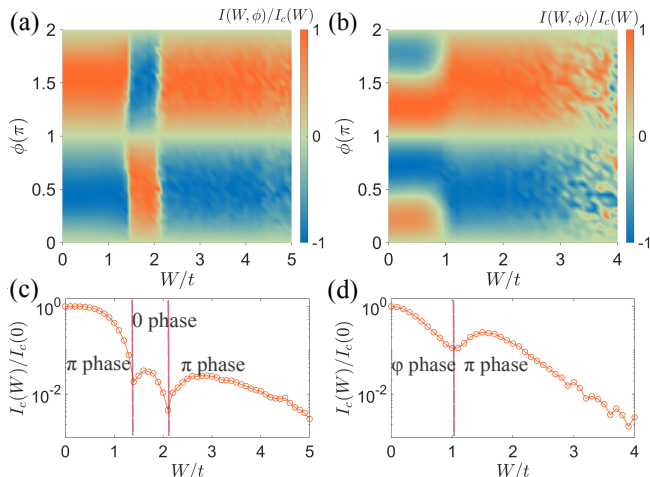


FIG. 5. (a-b) Evolution of the CPR with increasing disorder strength W for d_{xy} -wave AMs. In the clean limit, the AMJJ is in π phase for (a) with $L_x = 12$, and φ phase for (b) with $L_x = 22$. (c-d) The critical supercurrent as a function of W , corresponding to panels (a-b). Other parameters are $L_y = 30$, $V_b = 2t$, $t_J = 0.3t$, $\Delta = 0.1t$, $T = 0.05T_c$, and $\mu = -2t$. We have taken 5×10^3 disorder configurations for all plots.

to π phase again after passing through a narrow window of 0 phase, demonstrating a particular re-entrant phenomenon. In the d_{xy} -wave case, the φ phase remains fragile under disorder. As shown in Fig. 5(b), the φ phase undergoes a transition to π phase at a critical point $W_c \approx 1.0t$. The corresponding critical current curve has a minimum around the critical disorder strength as shown in Fig. 5(d). These consistent results confirm that disorder-induced phase transitions are generic features in the AMJJs even with different AM order orientations.

Conclusion and discussion.—In summary, we have demonstrated that disorder in AMJJs plays a crucial role: it induces phase transitions between distinct phases. In particular, disorder drives phase transition from the exotic π phase to conventional 0 phase even at a weak disorder strength, while the 0 to π phase transition is also enabled, establishing a reciprocal manner. Furthermore, we reveal that the anomalous φ phase is intrinsically fragile in presence of disorder. It is driven to either a 0 phase or π phase, depending on sign of the first harmonic component of the CPR. Our findings reveal the pronounced sensitivity of AMJJs to disorder such that the disorder effect should be considered for further functionalities of AMJJs. Given that many AM candidates are found to be metallic [63–69], our results are of direct experimental relevance.

The disorder-induced phase transitions of AMJJs are general phenomena even considering different types of disorder such as magnetic disorder [79]. Under the magnetic disorder, transitions between the π phase and the 0 phase also occur and the fragility of φ phase remains

[75]. In contrast, when the d -wave AM is replaced by odd-parity p -wave unconventional magnets, no phase transitions are observed [75]. It is interesting to further explore impact of the pairing amplitude symmetries in disordered AMJJs, such as the odd-parity pairing.

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- [75] See Supplemental Material for details of (S1) evolution of current-phase relations under disorder; (S2) Evolution of different order of Harmonics during the $0-\pi$ phase transition; (S3) general properties of AMJJs under general magnetic disorder; and (S4) properties of disordered p -wave magnets Josephson junctions, which includes Refs. [49, 50, 80, 81].
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Supplemental materials of “Disorder-Induced Phase Transitions in Altermagnetic Josephson Junctions”

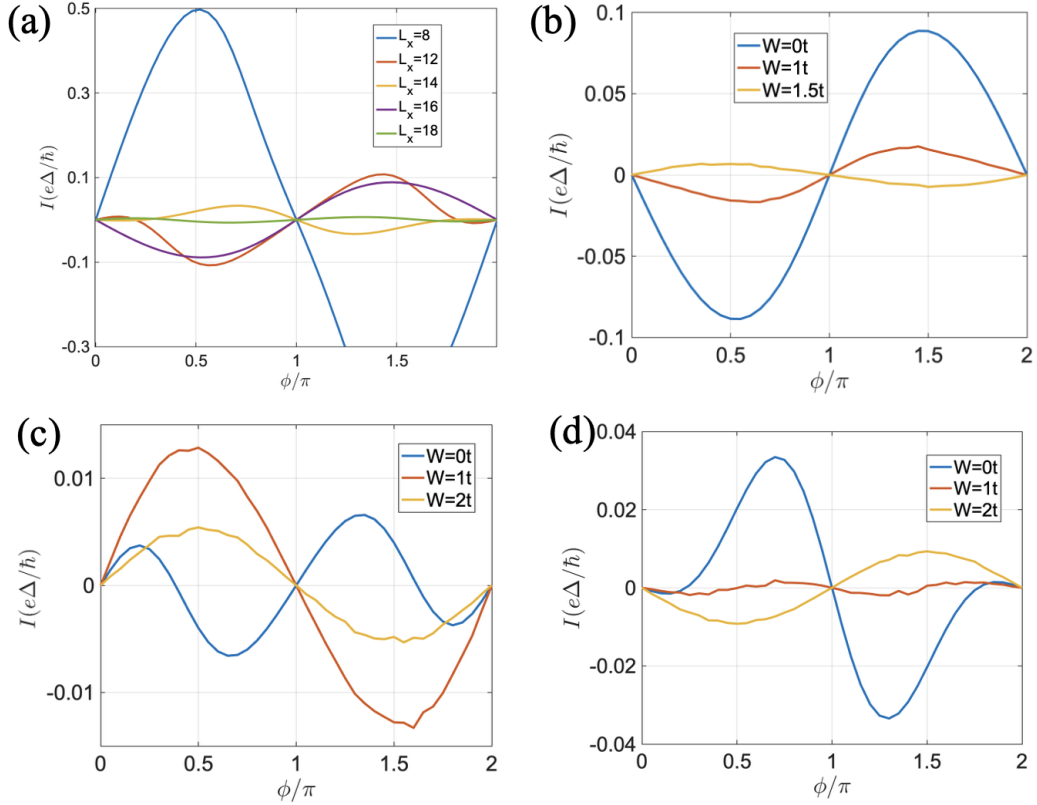


FIG. S1. (a) Plots of CPRs for different junction lengths. (b) Phase transitions from the exotic π phase to the conventional 0 phase with increasing disorder strength W . The junction length is fixed at $L_x = 16$. (c) Phase transitions from the anomalous φ phase to the conventional 0 phase as increasing W . The junction length is fixed at $L_x = 18$. (d) Phase transitions from the φ phase to the π phase as increasing W . The junction length is fixed at $L_x = 14$. Other parameters are $L_y = 30$, $V_b = 2t$, $t_J = 0.3t$, $\Delta = 0.1t$, $T = 0.05T_c$, and $\mu = -2t$. We have taken 5×10^3 disorder configurations for all plots.

Appendix S1: Evolution of current-phase relations under disorder

In this section, we present the current-phase relations (CPRs) of the disordered AMJJs before and after different types of phase transitions, corresponding to Fig. 2 in the main text. In the clean limit, the CPRs change as a function of the junction length L_x with a fixed width L_y . As shown in Fig. S1(a), the AMJJ exhibits the conventional 0 phase, the exotic π phase, and the anomalous φ phase at different junction lengths. Upon introducing disorder, the system undergoes phase transitions that modify the form of the CPR. Figures S1(b)-S1(d) demonstrate representative evolution of the CPR across such transitions. As demonstrated in Fig. S1(b), the system is in π phase at the clean limit with $W = 0$. With increasing the disorder strength W , the system is driven to the conventional 0 phase at a critical disorder strength $W \approx 1.2t$. Note that the CPR at $W = 1.5t$ clearly corresponds to the 0 phase.

For the φ phase, disorder can also induce a transition into the conventional 0 phase, as illustrated in Fig. S1(c). At weak disorder strength $W = 1.0t$, the AMJJ is already locating in a 0 phase and the critical current is even enhanced compared to the clean limit. Increasing W further, the critical current will be gradually suppressed as expected. Alternatively, the φ phase can be driven to π phase with different initial conditions, as shown in Fig. S1(d). In this case, the system experiences phase transition at around $W \approx 1.0t$ and it is well within the π phase for $W = 2.0t$. These results indicate that the anomalous φ phase is fragile in presence of disorder.

Appendix S2: Evolution of different order of harmonics during the 0- π phase transition

In this section, we analyze the evolution of harmonics components during the 0- π phase transition, complementing the results presented in Fig. 3 of the main text. As we have shown, the CRPs at different disorder strength are generally

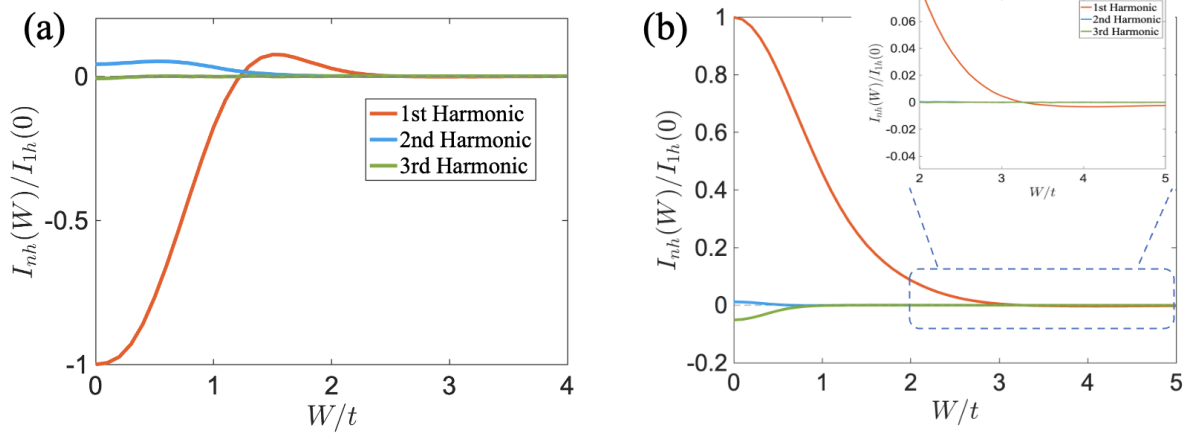


FIG. S2. (a) Different orders of harmonics in the CPRs as a function of disorder strength W , corresponding to phase transitions from the π phase to the 0 phase. (b) Similar to (a) but with the phase transitions from the 0 phase to the π phase. The zoomed inset plots the sign change of $I_{1h}(W)$. The data of panels (a) and (b) are from Figs. 2(a) and 2(b) of the main text, respectively.

expressed as $I(W, \phi) = \sum_{n=1}^{\infty} I_{nh}(W) \sin(n\phi)$ where $I_{nh}(W)$ denotes the amplitude of n -th order of harmonics. To extract the value of $I_{nh}(W)$, we perform Fourier transformation for the corresponding CPRs. We first focus on the phase transition from the exotic π phase to the 0 phase, as shown in Fig. S2 (a) [corresponding to Figs. 2(a) and 2(e) of the main text]. In the π phase, the CPR is dominated by the first harmonic I_{1h} with a negative value. The higher-order ones, such as the second and third orders, are vanishingly small compared to I_{1h} . With increasing disorder strength W , the higher-order harmonics $I_{2h}(W)$ and $I_{3h}(W)$ rapidly decay to zero. In contrast, the first harmonic $I_{1h}(W)$ (the absolute value) drops gradually and changes sign across the phase transition point, signaling the transition from the π phase to the 0 phase. Increasing W further, the first harmonic of $I_{1h}(W)$ will finally be suppressed close to zero, consistent with the substantially reduced critical current at the strong disorder strength.

A similar result is observed for the reversal transition from the convectional 0 phase to the exotic π phase [Fig. S2 (b)]. In the clean limit, the CPR is dominated by the first harmonic I_{1h} and the higher-order harmonic values I_{2h} and I_{3h} are vanishingly small. As increasing W , the first harmonic I_{1h} decays gradually and changes its sign across the phase transition point [see the zoomed region in Fig. S2(b)]. The system is transitioned to a π phase afterwards with significantly reduced critical current.

Appendix S3: General properties of AMJJs under general magnetic disorder

In this section, we demonstrate the disorder-induced phase transitions occur when more general type disorder is considered. To this end, we take the magnetic disorder of the form $V(\mathbf{r}) = \sum_{s=0,x,y} w_{\mathbf{r}}^s \sigma_s$ with spin-dependent random potentials at each lattice site $\mathbf{r}$. The random potential $w_{\mathbf{r}}^s$ takes value independently from the uniformly distributed interval $[-W/2, W/2]$ with W being the disorder strength. Here, the $w_{\mathbf{r}}^0 \sigma_0$ term indicates the usual nonmagnetic type disorder, and the $w_{\mathbf{r}}^x \sigma_x$ and $w_{\mathbf{r}}^y \sigma_y$ terms denote magnetic disorder that couples to the spin degrees of freedom. The main results are presented in Fig. S3. In Fig. S3(a), the initial π phase is driven to the 0 phase at a relatively weak disorder strength. Compared to the non-magnetic disorder case presented in Fig. S1(b), the critical disorder strength is reduced: the system is already in the 0 phase with $W = 1.0t$, whereas it remains in the π phase at the same disorder strength.

Figures S3(b) and S3(c) further present the transitions from φ phase to 0 phase and π phase, respectively. The overall behavior is qualitatively consistent with the non-magnetic disorder case presented in Figs. S1(c) and S1(d), but the values of critical disorder strength become smaller. This result is reasonable since extra randomness is introduced in the spin degrees of freedom, leading to a stronger decoherence effect.

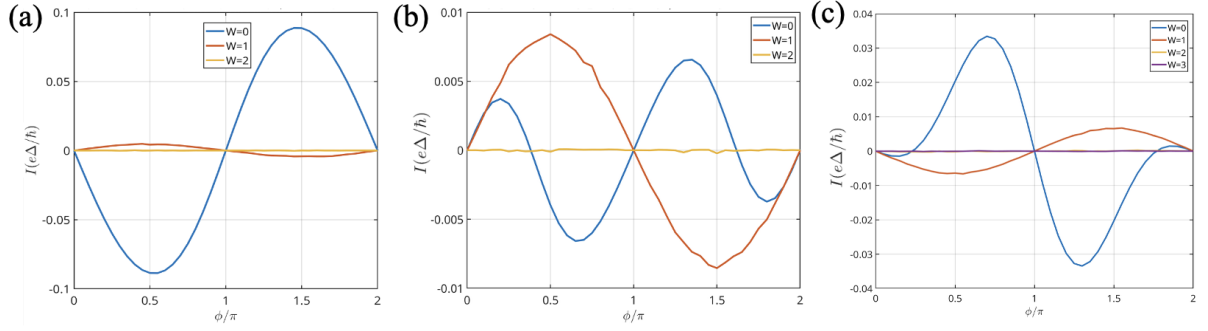


FIG. S3. Phase transitions of AMJJs under magnetic disorder. (a) Evolution of CPRs demonstrates phase transitions from the π phase to the 0 phase as increasing W . The junction length is fixed at $L_x = 16$. (b) Evolution of CPRs demonstrates phase transitions from the φ phase to the 0 phase. The junction length is fixed at $L_x = 18$. (c) Evolution of CPRs demonstrates phase transitions from the φ phase to the π phase. The junction length is fixed at $L_x = 14$. Other parameters are set at $L_y = 30$, $V_b = 2t$, $t_J = 0.3t$, $\Delta = 0.1t$, $T = 0.05T_c$, and $\mu = -2t$, the same as in the main text. We have taken 5×10^3 disorder configurations for all plots.

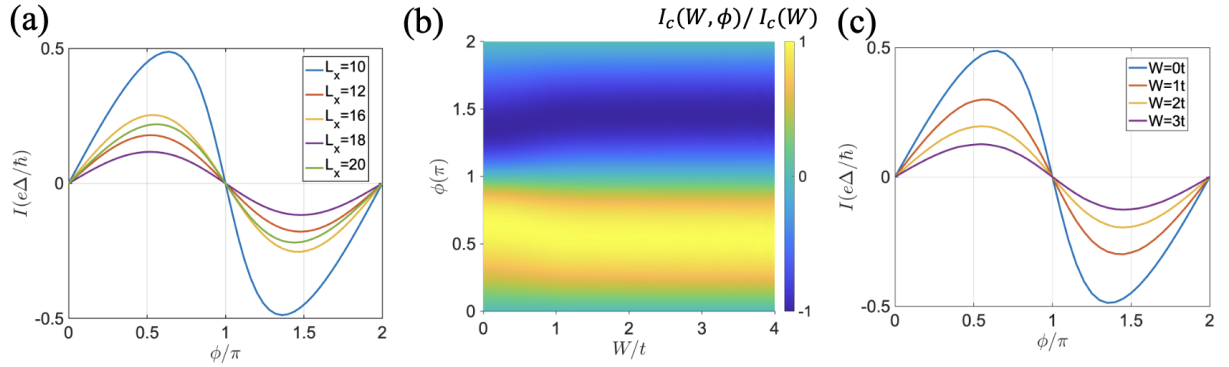


FIG. S4. (a) The CPRs in the p -wave magnets Josephson junctions as changing the junction length L_x . (b) The evolution of CPRs as a function of disorder strength W . Here we take $L_x = 10$. (c) The corresponding CPRs at different disorder strengths with a fixed junction length $L_x = 10$. Other parameters are set at $L_y = 30$, $V_b = 2t$, $p = 0.3t$, $\Delta = 0.1t$, $T = 0.05T_c$, and $\mu = -2t$. We have taken 2×10^3 disorder configurations for panels (b) and (c).

Appendix S4: Properties of disordered p -wave magnets Josephson junctions

In this section, we investigate the disordered p -wave magnets Josephson junctions. To this end, we employ the effective model for p -wave magnets as [80, 81]

$$H_p(\mathbf{k}) = t\mathbf{k}^2\sigma_0 + pk_x\sigma_x, \quad (\text{S4.1})$$

where t is the hopping amplitude and p is the magnitude of spin splitting. Note that the p -wave magnets do not break time reversal symmetry, and it has momentum-resolved spin polarization $\langle S_x \rangle$. Due to the odd-parity spin splitting, the system does not have inversion symmetry. To construct the corresponding tight-binding Hamiltonian on a 2D square lattice, we use the long-wavelength approximation that $k_x \approx \sin(k_x)$, $k_x^2 \approx 2(1 - \cos k_x)$. The junction geometry is the same as in Fig. 1(a) of the main text, with the d -wave AM being replaced by p -wave magnets.

The p -wave magnets Josephson junctions show similar properties to the conventional Josephson junctions [49, 50], i.e., the 0 phase with sinusoidal CPRs. As shown in Fig. S4(a), the CPR remains in the 0 phase for all junction lengths considered. As increasing the junction length L_x , the critical current decays gradually in an oscillating way, which is different from the conventional Josephson junctions. We next introduce the disorder of type $V(\mathbf{r}) = w_{\mathbf{r}}^0\sigma_0$ at each lattice site, where $w_{\mathbf{r}}^0$ is drawn from the uniformly distributed interval $[-W/2, W/2]$, with W denoting the disorder strength. As the disorder strength increases, the p -wave magnets Josephson junction remains in the 0 phase [see Fig. S4(b)]. Correspondingly, the critical current is reduced by disorder [see Fig. S4(c)]. In contrast to the d -wave AMJJs, no disorder-induced phase transition is observed.